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Cole–Hopf Solvers for Hamilton–Jacobi Reachability

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Viscous Hamilton–Jacobi (HJ) equations provide regularized value fields for backward reachability, but their nonlinear gradient terms complicate numerical integration. We apply a variable-coefficient Cole–Hopf transformation that maps the viscous HJ equation to a semilinear heat equation with a complete nonlinear source. Deterministic Fourier and finite-difference solvers integrate the transformed equation on periodic grids. A depth-first Monte Carlo Picard solver instead evaluates individual states through nested Gaussian value-and-gradient updates. On an analytic advection benchmark, increasing state dimension from 8 to 2048 at fixed sample count and Picard depth changes median point-query runtime from 0.0436 to 0.0760 seconds; the live coordinate payload is linear in dimension. Runtime grows from 0.0052 to 0.9355 seconds as samples per update increase from 4 to 48, so the experiment supports grid-free point-query storage rather than an end-to-end linear-runtime claim. Deterministic comparisons on rockets, relative-Dubins vehicles, a double integrator, and a starling-inspired model yield maximum sampled discrepancies below 3.3×10^{-3}. The $100{,}000$-state experiment queries one precomputed reduced field and is not a coupled population solve. Zero-sublevel shapes and topology counts remain numerical observations rather than continuum safety certificates.

Animations

Rockets comparison | Dubins comparison | Double integrator | Starling-inspired geometry | Monte Carlo scaling | Code

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1. Introduction

Modern complex systems — with their diverse interconnection of actuators, sensors, control systems, and the software modules that coordinate their various processes and events — require software verification tools capable of geometrically reasoning about their future behaviors. Given the continuous and/or hybrid systems characterization of their dynamics, simulation and testing, verification, observation, and further testing loop cycles are the standard norm for detecting unforeseen behavior ahead of deployment. Owing to their scale, however, analyzing system behavior based on closed-form analytic mathematical tools is often infeasible. This has stimulated efforts that complement traditional mathematical analysis tools from dynamical systems with emerging techniques in reachability [1], viability [2], or invariance [11] theories. These new tools allow exploring all possible scenarios where faults may arise for safety assurance; that is, ascertaining the freedom of a system from harm by generating evidence that a system, or any its components satisfy all specified requirements including functional and allocated baselines (i.e., *verification*) [8]. Examples of complex systems include smart manufacturing, ground [15] and air traffic management systems [25], flight control systems [18], or distributed systems.

Differential optimal control and game theories are useful formal verification tools for certifying the runtime safety assurance [11] of multiple agents interacting over a shared space in a *geometric* sense [13]. Both rely on a resolution of the Hamilton-Jacobi-Bellman (HJB) and/or the Hamilton-Jacobi-Isaacs (HJI) equation [7]. Being partial differential equations (PDEs), general high-dimensional HJB/HJI equations rarely admit closed-form solutions. Thus, approximation techniques have become the default mechanism for formal systems verification in these settings. Reachability theory [19, 25] is one such approximation technique. It entails computing the essentially non-oscillating (ENO) [24] Lax-Friedrichs [6] approximations to the HJ PDE via levelset methods [20–22]. Propagating the fronts of a conservation law on a mesh, the inviscid Euler equations' solution reduces to the derivative of the HJ problem's solution [10]; such that integrating this solution with total variation diminishing Runge-Kutta (TVD-RK) [23] produces generalized *viscosity solutions* to the original HJ problem.

HJ equation systems are employed within reachability analysis to evaluate a dynamical system's behavioral (or trajectory) evolution throughout a state space — for example, given its initial point sets (states). The problem is typically setup so that the initial set must satisfy *input or state constraints* in a *least restrictive sense* and under a *worst-possible disturbance* [16]. Reachability is thus a useful geometric reasoning tool for determining the states that can be reached from an initial condition in a finite number of steps [19, 21] . *Reachability* can be analyzed in a (i) *forward* sense, where system trajectories are examined to determine if they enter certain states from an *initial set*; (ii) *backward* sense, where system trajectories are examined to determine if they enter certain *target sets*; (iii) *reach set* sense, in which they are examined to see if states reach a *set* at a *particular time*; or (iv) *reach tube* sense, in which they are examined to determine that they reach a set *during a time interval*.

This article studies the Cauchy equations associated with terminal reachability. A terminal reachable set concerns target entry at a fixed horizon; a tube permits entry during an interval and requires a stopping operator. Reach–avoid problems additionally constrain the trajectory before target entry and require an obstacle formulation. The deterministic experiments solve fixed-horizon viscous Cauchy problems on tensor grids. The Monte Carlo experiment evaluates individual states without constructing such a grid. Neither experiment computes a coupled population game.

Consider the evolution-form of the Cauchy-type HJ PDE

$$\boldsymbol{v}_t + \boldsymbol{H}(t; \boldsymbol{x}, D\boldsymbol{v}) = 0 \text{ in } \Omega \times (0, T], \quad \boldsymbol{v}(0; \boldsymbol{x}) = \boldsymbol{g}(0; \boldsymbol{x}) \text{ on } \Omega \times \{t = 0\} \tag{HJ-IVP}$$

where $\boldsymbol{x} \in \Omega \subseteq \mathbb{R}^n$, the Hamiltonian $\boldsymbol{H}$ is continuous, and $D\boldsymbol{v}$ is the spatial gradient. The initial condition $\boldsymbol{g}$ is bounded and uniformly continuous. Crossing characteristics can destroy classical differentiability even when the Hamiltonian and initial condition are smooth [4]. Viscosity solutions provide generalized solutions. We study a fixed, positive-viscosity regularization before considering its approximation error.

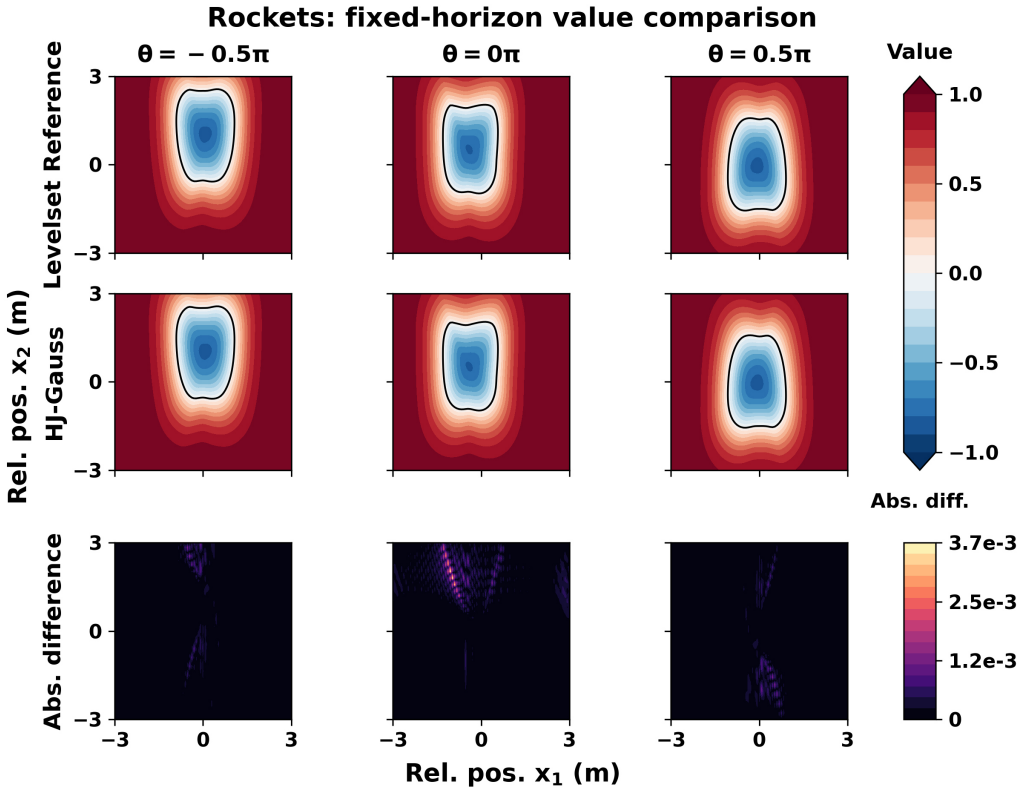


Figure 1: Normalized rockets benchmark at elapsed horizon $T = 0.5$ and viscosity $\delta = 0.08$. Rows show the `Levelset` reference [20, 21], the transformed solver, and their absolute difference. Columns select the headings $\theta = -\pi/2$, 0, and $\pi/2$. Both solvers use the same initial condition, Hamiltonian, and viscosity. Black contours mark the zero level set of each fixed-horizon value.

We study a variable-coefficient Cole–Hopf transformation of the viscous form of (HJ-IVP). Positive logarithmic inversion maps the equation to a semilinear heat equation while retaining the Hamiltonian and coefficient derivatives. The transformed identity supports both deterministic field integration and nonlinear Monte Carlo Picard point evaluation. Our results concern (i) the transformed source identity and logarithmic recovery formula; (ii) Fourier and finite-difference integration of the complete transformed equation; (iii) depth-first nested Gaussian evaluation of complete Picard value-and-gradient updates; (iv) analytic, refinement, sensitivity, and same-viscosity reference comparisons; and (v) measured point-query runtime and storage scaling with explicit claim limits.

The numerical study does not invoke the dimension-independent complexity results for multilevel Picard methods [12]. The nested implementation favors bounded live memory over runtime, which rises rapidly with sample count and Picard depth. Sampled residuals do not establish a continuum sign certificate. The $100{,}000$-state experiment interpolates one reduced relative-state field at independently sampled queries; population size affects the query count, not the dimension of the solved HJ equation.

Section 2 derives the transformed equation and describes both solver families. Section 3 reports scaling, analytic, and game-based numerical comparisons. Appendix A records the heat-kernel representation, while Appendix B states the scope of the convergence and residual arguments. Section 4 concludes the paper.

2. Cole–Hopf Transformation and Deterministic Solvers

Backward reachability associates a differential game with a Hamilton–Jacobi–Isaacs equation [19]. Let $\mathcal{U} \subset \mathbb{R}^m$ and $\mathcal{W} \subset \mathbb{R}^p$ be compact control sets. At physical time $s \in [0,T]$, let $\overline{\mathcal{U}}(s)$ and $\overline{\mathcal{W}}(s)$ contain the measurable control signals on $[s,T]$. Let $\mathcal{B}(s)$ contain the nonanticipative maps $\beta : \overline{\mathcal{U}}(s) \to \overline{\mathcal{W}}(s)$. The trajectory ξ starts in state $\boldsymbol{x}$ and satisfies

$$\dot{\xi}(r) = f(r, \xi(r), \boldsymbol{u}(r), \beta[\boldsymbol{u}](r)), \qquad \xi(s) = \boldsymbol{x}, \quad s \le r \le T, \tag{2.1}$$

where f is bounded, continuous and uniformly Lipschitz in $\boldsymbol{x}$. For bounded uniformly continuous running cost l and terminal cost $\boldsymbol{g}$, the lower value of the differential game is

$$\boldsymbol{v}_{\text{phys}}(s,\boldsymbol{x}) = \inf_{\beta \in \mathcal{B}(s)} \sup_{\boldsymbol{u} \in \overline{\mathcal{U}}(s)} \underbrace{\left\{ \int_s^T l(r, \xi(r), \boldsymbol{u}(r), \beta[\boldsymbol{u}](r))\, dr + \boldsymbol{g}(\xi(T)) \right\}}_{\Phi(s,\boldsymbol{x}) := \text{Payoff}}. \tag{2.2}$$

The target is represented by the zero sublevel set of $\boldsymbol{g}$. We assign $\boldsymbol{u}$ to the evader, which maximizes the payoff to avoid this set, and $\boldsymbol{w}$ to the pursuer, which minimizes the payoff to force target entry. The Hamiltonian is

$$\boldsymbol{H}_{\text{phys}}(s,\boldsymbol{x},\boldsymbol{p}) = \max_{\boldsymbol{u} \in \mathcal{U}} \min_{\boldsymbol{w} \in \mathcal{W}} \{\boldsymbol{p} \cdot f(s,\boldsymbol{x},\boldsymbol{u},\boldsymbol{w}) + l(s,\boldsymbol{x},\boldsymbol{u},\boldsymbol{w})\}. \tag{2.3}$$

For pure target-reach problems, the running cost is set to zero and the target is $\boldsymbol{g}$. The elapsed time $t = T - s$ converts the terminal condition into an initial condition so that if we write $\boldsymbol{v}(t,\boldsymbol{x}) = \boldsymbol{v}_{\text{phys}}(T-t,\boldsymbol{x})$ and define $\boldsymbol{H}(t,\boldsymbol{x},\boldsymbol{p}) = -\boldsymbol{H}_{\text{phys}}(T-t,\boldsymbol{x},\boldsymbol{p})$, the elapsed equation is $\boldsymbol{v}_t + \boldsymbol{H}(t,\boldsymbol{x},D\boldsymbol{v}) = 0$. Fixing $\delta > 0$ gives the viscous parabolic regularization

$$\boldsymbol{v}_t^\delta + \boldsymbol{H}(t,\boldsymbol{x},D\boldsymbol{v}^\delta) = \frac{\delta}{2}\Delta \boldsymbol{v}^\delta, \qquad \boldsymbol{v}^\delta(0,\boldsymbol{x}) = \boldsymbol{g}(\boldsymbol{x}). \tag{2.4}$$

Equation (2.4) supplies approximation results under the Hamiltonian and initial condition [4, 5]. The transformation below concerns classical solutions of this regularized equation.

(a) HJ Linearization

Write $\boldsymbol{v} = \boldsymbol{v}^\delta$ within this subsection and choose a prescribed coefficient $\boldsymbol{c} \in C_b^{1,2}([0,T] \times \mathbb{R}^n)$ with $0 < \boldsymbol{c}_- \le \boldsymbol{c} \le \boldsymbol{c}_+$. Define the transformed value and its initial condition by

$$\boldsymbol{\omega} = e^{-\boldsymbol{c}\boldsymbol{v}}, \qquad \boldsymbol{\omega}_0(\boldsymbol{x}) = e^{-\boldsymbol{c}(0,\boldsymbol{x})\boldsymbol{g}(\boldsymbol{x})}. \tag{2.5}$$

For a scalar $z > 0$ and vector $q \in \mathbb{R}^n$, introduce

$$V(t,\boldsymbol{x},z) = -\frac{\log z}{\boldsymbol{c}(t,\boldsymbol{x})}, \qquad P(t,\boldsymbol{x},z,q) = -\frac{q}{\boldsymbol{c}z} + \frac{\log z}{\boldsymbol{c}^2} D\boldsymbol{c}, \tag{2.6}$$

$$\boldsymbol{R}_{\text{alg}}[\boldsymbol{c},V,P] = \boldsymbol{c}\left(\boldsymbol{H}(t,\boldsymbol{x},P) - \frac{\delta \boldsymbol{c}}{2}|P|^2\right), \tag{2.7}$$

$$\boldsymbol{R}_{\text{der}}[\boldsymbol{c},V,P] = -\boldsymbol{c}_t V + \frac{\delta}{2}(\Delta \boldsymbol{c})V - \frac{\delta}{2}|D\boldsymbol{c}|^2 V^2 + \delta(1 - \boldsymbol{c}V) D\boldsymbol{c} \cdot P, \tag{2.8}$$

$$F(t,\boldsymbol{x},z,q) = z\{\boldsymbol{R}_{\text{alg}}[\boldsymbol{c},V,P] + \boldsymbol{R}_{\text{der}}[\boldsymbol{c},V,P]\}. \tag{2.9}$$

Then, the transformed equation is

$$\boldsymbol{\omega}_t - \frac{\delta}{2}\Delta \boldsymbol{\omega} = F(t,\boldsymbol{x},\boldsymbol{\omega},D\boldsymbol{\omega}), \qquad \boldsymbol{\omega}(0) = \boldsymbol{\omega}_0. \tag{2.10}$$

For positive solutions, logarithmic inversion recovers the original viscous equation — reducing to a heat equation when the sum in (2.9) vanishes along the solution. Let S_t denote the heat semigroup

Algorithm 1 Deterministic integration of the transformed equation

1: Choose $\delta > 0$, a prescribed smooth coefficient $\boldsymbol{c} \geq \boldsymbol{c}_- > 0$, a periodic box, and a time step Δt.
2: Discretize the box by Fourier collocation or centered finite differences.
3: Set $z^0 = \exp(-\boldsymbol{c}(0,\cdot)\boldsymbol{g})$.
4: **for** $j = 0, 1, \ldots, J-1$ **do**
5: Reconstruct Dz^j and evaluate the complete source $F(t_j, \cdot, z^j, Dz^j)$.
6: Advance diffusion and source terms by exponential Runge–Kutta or classical Runge–Kutta integration.
7: Reject the step if the transformed field is nonfinite or nonpositive.
8: **end for**
9: Recover the value from $\boldsymbol{v}^J = -\boldsymbol{c}(t_J, \cdot)^{-1} \log z^J$.

with covariance $\delta t I_n$. The transformed equation admits the *mild representation*

$$\boldsymbol{\omega}(t) = S_t \boldsymbol{\omega}_0 + \int_0^t S_{t-s} F(s, \cdot, \boldsymbol{\omega}(s), D\boldsymbol{\omega}(s))\, ds, \tag{2.11}$$

which couples values and gradients. Each mild update substitutes the preceding iterate into both source arguments.

For $f_k(s, \boldsymbol{x}) = F(s, \boldsymbol{x}, z_k, Dz_k)$, the value and gradient update as an ordered pair:

$$(z_{k+1}, Dz_{k+1})(t) = (S_t, DS_t)\boldsymbol{\omega}_0 + \int_0^t (S_{t-s}, DS_{t-s}) f_k(s)\, ds. \tag{2.12}$$

$$\boldsymbol{v}_{k+1}(t, \boldsymbol{x}) = -\frac{\log z_{k+1}(t, \boldsymbol{x})}{\boldsymbol{c}(t, \boldsymbol{x})}, \qquad D\boldsymbol{v}_{k+1} = -\frac{Dz_{k+1}}{\boldsymbol{c} z_{k+1}} - \frac{\boldsymbol{v}_{k+1}}{\boldsymbol{c}} D\boldsymbol{c}. \tag{2.13}$$

Appendix A derives the Gaussian forms of the source integral and coefficients. We use two numerical realizations of these formulas. Deterministic solvers integrate the transformed field on periodic grids. A separate depth-first Monte Carlo solver evaluates the Picard pair in (2.12) at individual states.

(b) Fixed-horizon sets and reachable tubes

Let the target be the zero sublevel set

$$\mathcal{L}_0 = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{g}(\boldsymbol{x}) \leq 0\}. \tag{2.14}$$

Under the value convention of (2.2), the zero sublevel set of the fixed-horizon solution identifies states from which the pursuer can force terminal target entry against the evader. The elapsed-time Cauchy equation is (2.4). This article computes that fixed-horizon field.

A backward reachable tube permits target entry before the terminal horizon. For an autonomous target-reach problem, its inviscid value satisfies a stopping equation of the form

$$\partial_t \boldsymbol{v}_{\mathrm{tube}} + \min\{0, \boldsymbol{H}(\boldsymbol{x}, D\boldsymbol{v}_{\mathrm{tube}})\} = 0, \qquad \boldsymbol{v}_{\mathrm{tube}}(0) = \boldsymbol{g}. \tag{2.15}$$

Reach–avoid problems require an additional obstacle condition [19]. The stopping and obstacle equations are not used in the numerical comparisons below. We therefore reserve "tube" for this distinct formulation and call the computed objects fixed-horizon reachable sets.

Write $\boldsymbol{v} = \boldsymbol{v}^\delta$ in the transformation results that follow.

Lemma 2.1 (Transformation Identity). *Suppose that $\boldsymbol{v}(t; \boldsymbol{x}), \boldsymbol{c}(t; \boldsymbol{x}) \in C^{1,2}$ and set $\boldsymbol{\omega} = e^{-\boldsymbol{c}\boldsymbol{v}}$. Define the residual $r_v = \boldsymbol{v}_t + \boldsymbol{H}(t, \boldsymbol{x}, D\boldsymbol{v}) - \delta \Delta \boldsymbol{v}/2$. Then*

$$\boldsymbol{\omega}_t - \frac{\delta}{2} \Delta \boldsymbol{\omega} = \boldsymbol{\omega}\{\boldsymbol{R}_{\mathrm{alg}}[\boldsymbol{c}, \boldsymbol{v}, D\boldsymbol{v}] + \boldsymbol{R}_{\mathrm{der}}[\boldsymbol{c}, \boldsymbol{v}, D\boldsymbol{v}] - \boldsymbol{c} r_v\}. \tag{2.16}$$

Proof. Set $\phi = \boldsymbol{c}\boldsymbol{v}$. Differentiation gives $\boldsymbol{\omega}_t = -\boldsymbol{\omega}\phi_t$, $D\boldsymbol{\omega} = -\boldsymbol{\omega}D\phi$, and $\Delta\boldsymbol{\omega} = \boldsymbol{\omega}(|D\phi|^2 - \Delta\phi)$. Expand $D\phi = \boldsymbol{c}D\boldsymbol{v} + \boldsymbol{v}D\boldsymbol{c}$ and $\Delta\phi = \boldsymbol{c}\Delta\boldsymbol{v} + 2D\boldsymbol{c}\cdot D\boldsymbol{v} + \boldsymbol{v}\Delta\boldsymbol{c}$. Substitution cancels the two terms containing $\boldsymbol{c}\Delta\boldsymbol{v}$. The remaining terms give (2.16). Since $\boldsymbol{c}\boldsymbol{\omega} > 0$, the original residual vanishes if and only if the transformed residual vanishes. □

Corollary 2.1 (Homogeneous Heat Flow Reduction)**.** *Considering* (2.16)*, homogeneous heat flow holds if and only if* $\boldsymbol{R}_{\mathrm{alg}} + \boldsymbol{R}_{\mathrm{der}} = 0$*. A sufficient condition is a positive constant* $\boldsymbol{c}_0 > 0$ *and* $\boldsymbol{H}(t, \boldsymbol{x}, D\boldsymbol{v}) := \delta\boldsymbol{c}_0|D\boldsymbol{v}|^2/2$*. The Hamiltonian* $\boldsymbol{H}(p) = |p|^2/2$ *satisfies this condition whenever* $\boldsymbol{c}_0 = 1/\delta$*.*

Proof. Apply Lemma 2.1 with $r_v = 0$. A constant $\boldsymbol{c}$ eliminates the derivative residual, and the quadratic relation eliminates the algebraic residual. □

Corollary 2.2 (Source representation and value recovery)**.** *Suppose* $\boldsymbol{\omega}_0$ *is bounded and uniformly continuous, and the source* $f(s, \boldsymbol{x}) = F(s, \boldsymbol{x}, \boldsymbol{\omega}(s, \boldsymbol{x}), D\boldsymbol{\omega}(s, \boldsymbol{x}))$ *is bounded and continuous. Every bounded classical solution satisfies*

$$\boldsymbol{\omega}(t, \boldsymbol{x}) = \mathbb{E}[\boldsymbol{\omega}_0(\boldsymbol{x} + \sqrt{\delta t}\, Z)] + \int_0^t \mathbb{E}[f(s, \boldsymbol{x} + \sqrt{\delta(t-s)}\, Z)]\, ds, \tag{2.17a}$$

$$\boldsymbol{v}(t, \boldsymbol{x}) = -\frac{1}{\boldsymbol{c}(t, \boldsymbol{x})} \log \boldsymbol{\omega}(t, \boldsymbol{x}), \qquad Z \sim \mathcal{N}(0, I_n). \tag{2.17b}$$

Proof. Variation of constants for the heat equation gives the mild representation. The whole-space heat kernel has unit mass and equals the indicated Gaussian density. Invert the transformation at the evaluation point. □

The source contains the unknown solution. If $|\boldsymbol{v}| \le G$, then $e^{-\boldsymbol{c}_+ G} \le \boldsymbol{\omega} \le e^{\boldsymbol{c}_+ G}$.

Proposition 2.1 (Gradient for `HJ-Gauss`)**.** *Under the preceding bounded-source hypotheses, for* $t > 0$ *the gradient satisfies*

$$D\boldsymbol{\omega}(t, \boldsymbol{x}) = \frac{1}{\sqrt{\delta t}} \mathbb{E}[Z\boldsymbol{\omega}_0(\boldsymbol{x} + \sqrt{\delta t}\, Z)] + \int_0^t \frac{\mathbb{E}[Z f(s, \boldsymbol{x} + \sqrt{\delta(t-s)}\, Z)]}{\sqrt{\delta(t-s)}}\, ds, \tag{2.18a}$$

$$D\boldsymbol{v}(t, \boldsymbol{x}) = -\frac{D\boldsymbol{\omega}(t, \boldsymbol{x})}{\boldsymbol{c}(t, \boldsymbol{x})\boldsymbol{\omega}(t, \boldsymbol{x})} - \frac{\boldsymbol{v}(t, \boldsymbol{x})}{\boldsymbol{c}(t, \boldsymbol{x})} D\boldsymbol{c}(t, \boldsymbol{x}). \tag{2.18b}$$

Proof. Differentiate the Gaussian kernel with respect to its center. Its derivative has integrable L^1 norm bounded by $\sqrt{n/(\delta(t-s))}$. This bound justifies differentiation of the source integral. The product rule applied to $\boldsymbol{v} = -\boldsymbol{c}^{-1}\log\boldsymbol{\omega}$ gives the second formula. □

(c) Deterministic discretizations

The Fourier solver represents the transformed field on a periodic tensor grid. Spectral differentiation supplies $D\boldsymbol{\omega}$, and a two-stage exponential Runge–Kutta update integrates diffusion and the complete nonlinear source. A one-third collocation cutoff suppresses unresolved source modes. The finite-difference solver instead uses fourth-order centered derivatives and classical fourth-order Runge–Kutta integration. Neither solver clips the transformed field; a nonpositive iterate terminates the computation.

The reference solver evolves the original viscous value equation. It uses fourth-order centered differences for the spectral comparisons and sixth-order centered differences for the finite-difference comparisons. These independent discretizations share the Hamiltonian, initial condition, viscosity, horizon, and observation queries. Their discrepancies measure agreement between the stated numerical schemes, not error relative to an unknown continuum solution.

Algorithm 2 Depth-first nonlinear Monte Carlo Picard point evaluation

Choose a point $(t, \boldsymbol{x})$, Picard level K, positive viscosity δ, and an even sample count N.
Estimate the heat value and score gradient with $N/2$ antithetic Gaussian pairs.
if $K = 0$ **then**
    Return the heat value and gradient.
end if
Draw $N/2$ independent source times and Gaussian vectors.
for each antithetic pair **do**
    Recursively evaluate level $K - 1$ at both displaced source states.
    Evaluate the complete nonlinear source $F(s, \boldsymbol{x}, z_{K-1}, Dz_{K-1})$ at both states.
end for
Add the importance-weighted source value and score-gradient estimates to the heat pair.
Reject a nonfinite or nonpositive transformed estimate; otherwise recover $(\boldsymbol{v}_K, D\boldsymbol{v}_K)$ by (2.13).

(d) Depth-first nonlinear Monte Carlo Picard evaluation

The stochastic solver evaluates one Picard iterate at a requested state without reconstructing a tensor-product field. Level zero estimates $(S_t, DS_t)\boldsymbol{\omega}_0$ with antithetic Gaussian pairs. At each higher level, the solver draws a source time $S = t(1 - U^2)$ with $U \sim \mathcal{U}(0, 1)$ and antithetic displacements $\pm\sqrt{\delta(t - S)}Z$. It recursively evaluates (z_{k-1}, Dz_{k-1}) at both displaced states, forms the complete source in (2.9), and accumulates the Duhamel value and score-gradient estimates. The source-time density is proportional to $(t - s)^{-1/2}$; its reciprocal cancels the terminal-time singularity in the score estimator.

Depth-first recursion releases completed branches before evaluating the next branch. For fixed Picard level, its counted live coordinate payload is linear in sample count and state dimension. The work tree nevertheless grows rapidly with both sample count and level. This implementation therefore tests grid-free point-query storage, not dimension-independent or linear end-to-end runtime. It also provides no field between queried states unless a separate reconstruction method is supplied.

The mild map also supplies a local analytical interpretation of the transformed equation.

Theorem 2.1 (Conditional finite-interval contraction)**.** *Let $X_\tau = C([0, \tau]; \mathrm{BUC}^1)$ carry the sum of the uniform value and gradient norms. Suppose a nonempty closed set $\mathcal{A} \subset X_\tau$ is invariant under the mild map, its functions satisfy $a \le z \le b$ and $|Dz| \le Q$ with $a > 0$, and the bounded source is uniformly Lipschitz in (z, Dz) with constant L. If*

$$\theta = L\left(\tau + 2\sqrt{n\tau/\delta}\right) < 1, \tag{2.19}$$

then the mild map has a unique fixed point in $\mathcal{A}$. If that fixed point is classical, its logarithmic inverse solves the viscous Hamilton–Jacobi equation.

Appendix B proves the contraction statement. The scaling experiment runs finite Picard levels and does not verify the theorem's invariant-set hypotheses or convergence to the fixed point.

(e) Residual and viscosity approximation errors

For a bounded smooth approximation U with bounded residual

$$r_U = U_t + \boldsymbol{H}(t, \boldsymbol{x}, DU) - \frac{\delta}{2}\Delta U, \tag{2.20}$$

comparison gives a value-error bound by the initial-condition error plus $\int_0^t \|r_U(s)\|_\infty ds$. If an independent vanishing-viscosity theorem also gives a bound $C_{\mathrm{vis}}\sqrt{\delta}$, the combined error bound is

$$E(t) = \|U(0) - \boldsymbol{g}\|_\infty + \int_0^t \|r_U(s)\|_\infty ds + C_{\mathrm{vis}}\sqrt{\delta}. \tag{2.21}$$

The transformed residual satisfies $d = -\boldsymbol{c}\exp(-\boldsymbol{c}U)r_U$ (see Appendix B).

Corollary 2.3 (Conditional sign certification)**.** *Suppose $|U - \boldsymbol{v}^0| \le E$ on a specified set, with $\boldsymbol{v}^0 \le 0$ denoting the target-reaching class. Then $U > E$ implies $\boldsymbol{v}^0 > 0$, and $U < -E$ implies $\boldsymbol{v}^0 < 0$. The remaining states form the abstention band $|U| \le E$.*

The proof is given in Appendix B.

3. Numerical Validation and Scaling

The experiments test three distinct properties: point-query scaling of the nonlinear Monte Carlo Picard solver, discretization accuracy of the deterministic transformed solvers, and repeated evaluation of a precomputed field. The deterministic comparisons share the viscosity, Hamiltonian, bounded initial condition, and horizon. Their discrepancy therefore excludes viscosity bias but retains the numerical errors of both solvers. All recomputed fields use the elapsed-time convention of Section 2. The inherited raster figures label the comparison field "Ground Truth." In this manuscript, that label denotes the independent numerical reference and not the unknown continuum solution.

(a) Nonlinear Monte Carlo Picard scaling

The scaling benchmark uses the bounded initial value $g(\boldsymbol{x}) = 0.1n^{-1}\sum_{j=1}^n \cos x_j$, viscosity $\delta = 0.4$, and a linear-advection Hamiltonian $\boldsymbol{H}(p) = a \cdot p$. The velocity components range from $0.05/\sqrt{n}$ to $0.15/\sqrt{n}$. The corresponding viscous HJ solution is available analytically, while its Cole–Hopf source remains nonlinear. We evaluate the zero state at horizon $t = 0.15$. Each reported runtime and memory value is the median of 3 fresh-process runs.

At Picard level 2 and 12 samples per update, increasing state dimension from 8 to 2048 changes median runtime from 0.0436 to 0.0760 seconds. The fitted log–log runtime slope is 0.085, and the maximum absolute value error over this sweep is 5.53×10^{-4}. The measured Python-traced-memory slope is 0.694; fixed interpreter costs dominate the smaller cases, while the directly counted coordinate payload is linear in dimension.

The sample and depth sweeps expose the cost hidden by the dimension result. At dimension eight and Picard level two, increasing samples per update from 4 to 48 increases median runtime from 0.0052 to 0.9355 seconds, with fitted log–log slope 2.05. With six samples, increasing the Picard level from 0 to 3 increases runtime from 0.00062 to 0.05399 seconds. Figure 2 therefore supports a grid-free point-query storage claim. It does not support an $O(Nn)$ end-to-end runtime claim, favorable scaling in Picard depth, or reconstruction of a continuum value field.

(b) Starling-inspired relative-state geometry

The starling-inspired case tests repeated evaluation of a controlled relative-state field. It does not model a coupled flock trajectory. The solved state is $\boldsymbol{x} = (x_1, x_2, \theta)$, comprising two relative positions and one relative heading. Equal vertical control bounds cancel in the four-component physical Hamiltonian, while an altitude-independent terminal payoff gives $p_z = 0$. The four-component query state can therefore be evaluated through the reduced three-dimensional field.

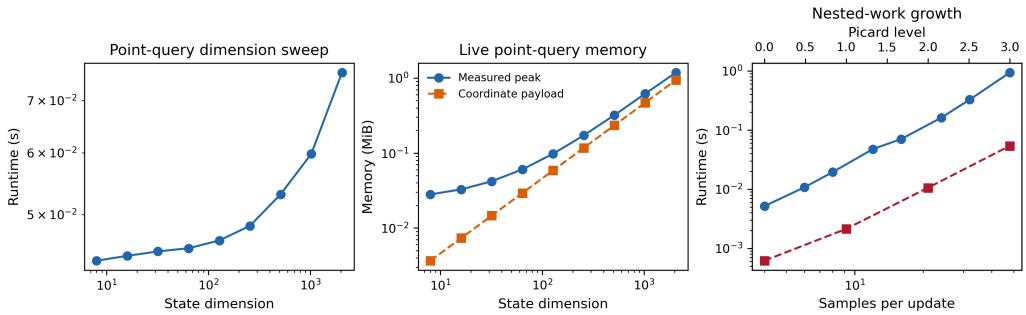


Figure 2: Scaling of one nonlinear Monte Carlo Picard point query. Left: runtime at fixed sample count and Picard level. Center: Python-traced peak memory and the counted depth-linear coordinate payload. Right: runtime growth with samples per update and Picard level. Markers are medians of three fresh-process runs; connecting lines guide the eye.

For unit vehicle speeds and steering bounds, the elapsed-time Hamiltonian is

$$\boldsymbol{H}(\boldsymbol{x}, p) = p_1(1 - \cos\theta) - p_2 \sin\theta - |p_1 x_2 - p_2 x_1 - p_3| + |p_3|. \tag{3.1}$$

The baseline terminal payoff is the minimum of seven smoothed radial payoffs. Their centers lie on a ring of radius 1.15, and each target disk has radius 0.6. Taking the minimum forms the union of the seven terminal targets before evolution. This construction does not assert that independently optimized flock values commute with shared controls.

Figure 3 shows three terminal-payoff configurations used to visualize annular, expanding, and perforated zero-sublevel geometries. The labels connect the shapes to starling-inspired behavior. They are geometric descriptions, not biological or game-theoretic classification theorems.

(i) Finite-window topology diagnostics

For a two-dimensional slice, let $n_{\mathrm{comp}}(t)$ be the number of connected components of the numerical zero sublevel set and let $\beta_1(t)$ count enclosed complementary components. The finite-window Euler diagnostic is

$$\chi(t) = n_{\mathrm{comp}}(t) - \beta_1(t). \tag{3.2}$$

The implementation thresholds each saved slice, labels its connected components, and counts enclosed holes. Figure 4 reports these integer diagnostics for both numerical fields. Agreement of the two discretizations supports the displayed finite-grid geometry. It does not locate a continuum topology transition or prove a barrier certificate.

(ii) Population query experiment

A seeded cohort contains $100{,}000$ independently sampled states. Horizontal positions are uniform on $[-3, 3]^2$, altitude is uniform on $[-1, 1]$, and heading is uniform on $[-\pi, \pi)$. Each query uses its horizontal position and heading to interpolate the same precomputed three-dimensional field; altitude is retained in the cohort record but does not enter the reduced solution. The experiment therefore measures the cost and consistency of repeated interpolation. It is not a joint $400{,}000$-dimensional solve, a simulation of interacting birds, or a population-wide safety certificate.

Figure 5 displays $2{,}000$ cohort states for legibility. The overlaid contour is the $\theta = 0$ slice, while each stored value uses the corresponding state's heading.

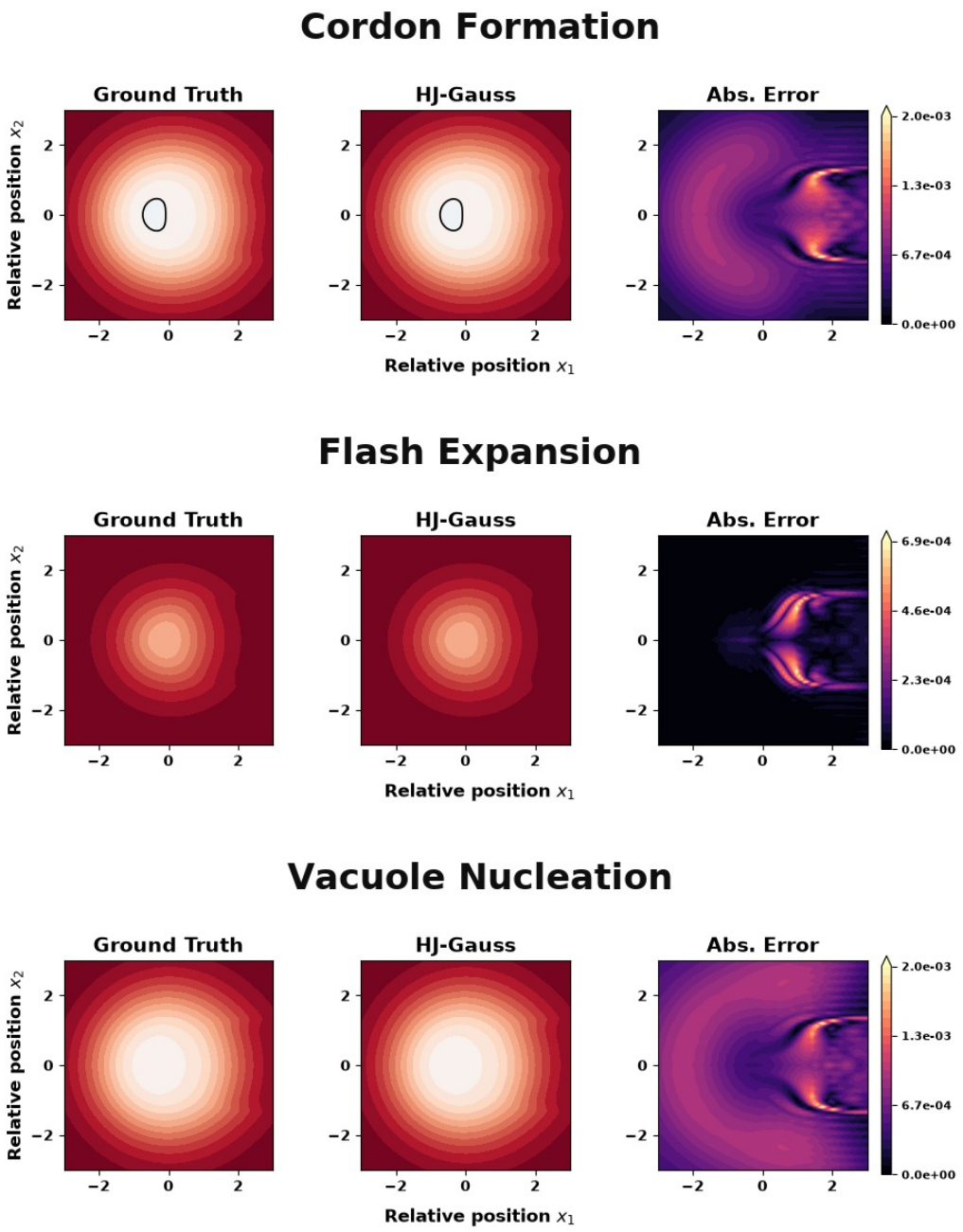


Figure 3: Starling-inspired fixed-horizon values for three terminal-payoff configurations. Each group shows the reference field, transformed-solver field, and absolute difference. Black contours mark the zero level sets.

(c) The Double Integrator Time-optimal Control Problem

The double-integrator benchmark asks which states can reach the origin under a bounded control [3, 26]. With second-order dynamics $\ddot{\boldsymbol{x}} = \boldsymbol{u}$, the first-order state satisfies

$$\dot{\boldsymbol{x}}_1(t) = \boldsymbol{x}_2(t), \qquad \dot{\boldsymbol{x}}_2(t) = \boldsymbol{u}(t), \qquad |\boldsymbol{u}(t)| \leq 1. \tag{3.3}$$

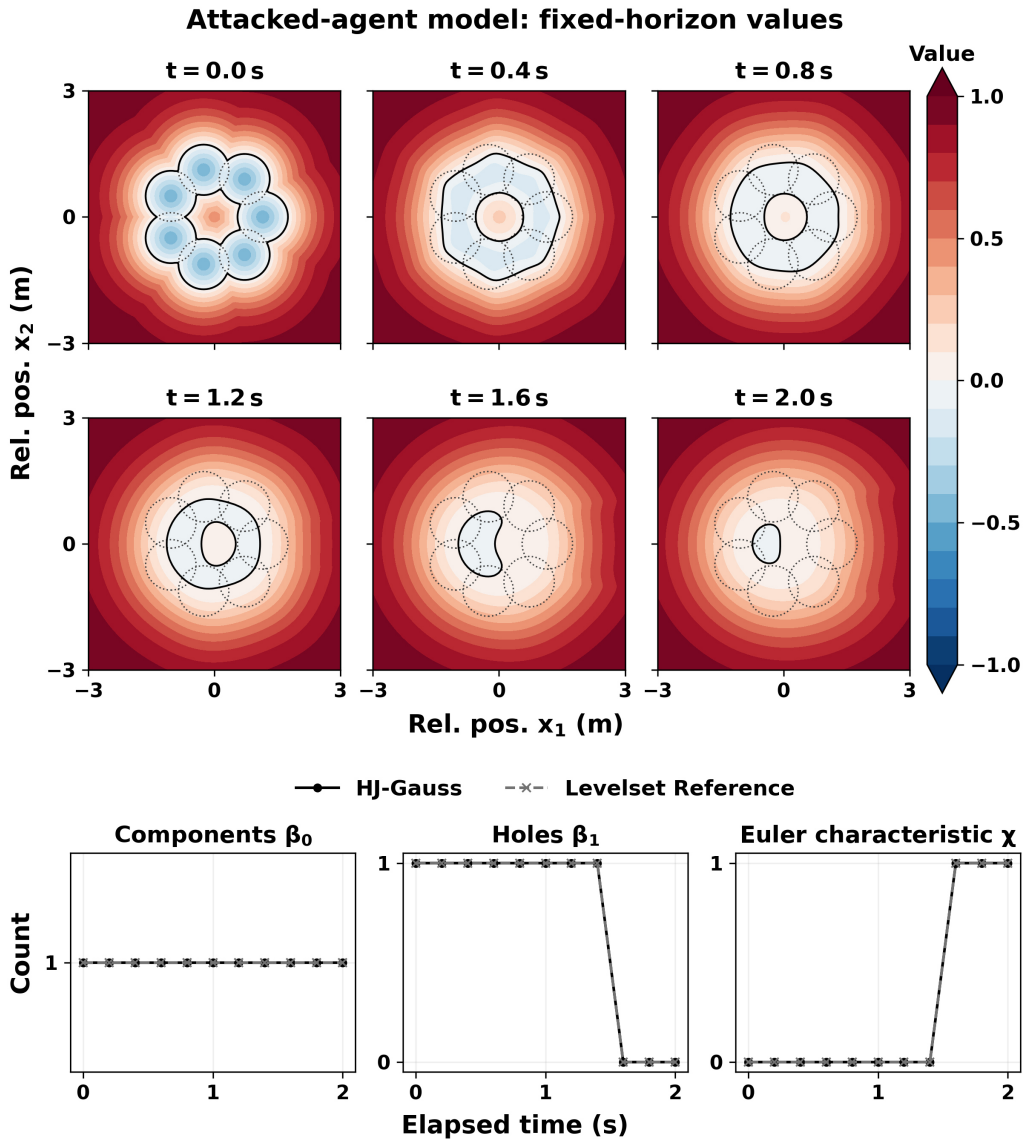


Figure 4: Finite-window geometry of the starling-inspired value. Upper panels show six elapsed horizons, black zero contours, and dotted terminal disks. The lower panel reports component, hole, and Euler counts from thresholded numerical slices.

The physical-time Hamiltonian is $\boldsymbol{H}_{\text{phys}} = p_1\boldsymbol{x}_2 + p_2\boldsymbol{u}$. Minimization over the bounded control gives $\boldsymbol{u}^* = -\operatorname{sign}(p_2)$ for $p_2 \neq 0$. The elapsed-time Hamiltonian used by both numerical solvers is

$$\boldsymbol{H}(\boldsymbol{x}, p) = -p_1\boldsymbol{x}_2 + |p_2|. \tag{3.4}$$

Figure 6 compares the transformed solution with the analytical time-to-origin geometry [21]. The bottom row shows the absolute field discrepancy at each saved time. Table 2 reports a maximum sampled reference discrepancy of 5.60×10^{-4}; this field comparison does not establish set containment.

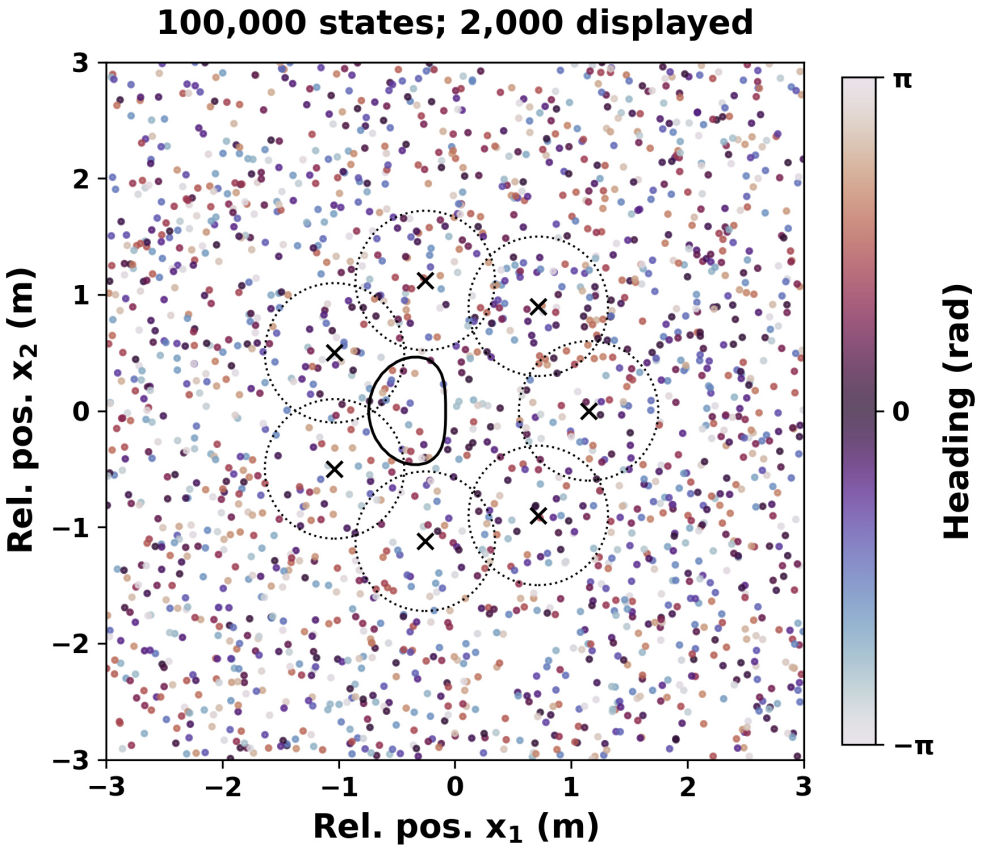


Figure 5: Planar projection of 2,000 states from the 100,000-state query cohort. Color denotes heading. Crosses and dotted circles mark the seven target centers and terminal disks; the black contour is the $\theta = 0$ slice.

(d) The Rockets Launch Problem on a Plane Benchmark

The planar rocket example of Dreyfus [9], recast as a differential game in [21], provides the next benchmark. Its relative dynamics are

$$\dot{x}_1 = a\cos\theta + u_e x_1, \quad \dot{x}_2 = a\sin\theta + a + u_p x_1 - g, \quad \dot{\theta} = u_p - u_e. \tag{3.5}$$

The scalar g is gravitational acceleration, a is vehicle acceleration, and $\theta \in \mathbb{S}^1$ is the relative heading. The attitude-rate controls satisfy $u_e, u_p \in [-1, 1]$. The evader maximizes the terminal payoff, while the pursuer minimizes it. The elapsed-time Hamiltonian is [21]

$$\boldsymbol{H}(\boldsymbol{x}, p) = -ap_1\cos\theta - p_2(g - a - a\sin\theta) - |p_1 x_1 - p_3| + |p_2 x_1 + p_3|, \tag{3.6}$$

The experiment sets $a = 1$, $g = 2$, capture radius 1.5, viscosity $\delta = 0.08$, and horizon $T = 0.5$. Figure 1 compares the transformed solver with the `LevelSetPy` reference [20, 21].

The terminal payoff is a smoothed signed-distance field for a cylindrical capture target. The three displayed headings use a shared color scale and observation window. Table 2 reports the maximum sampled field discrepancy, while Table 1 reports time-step, coefficient, and box sensitivity.

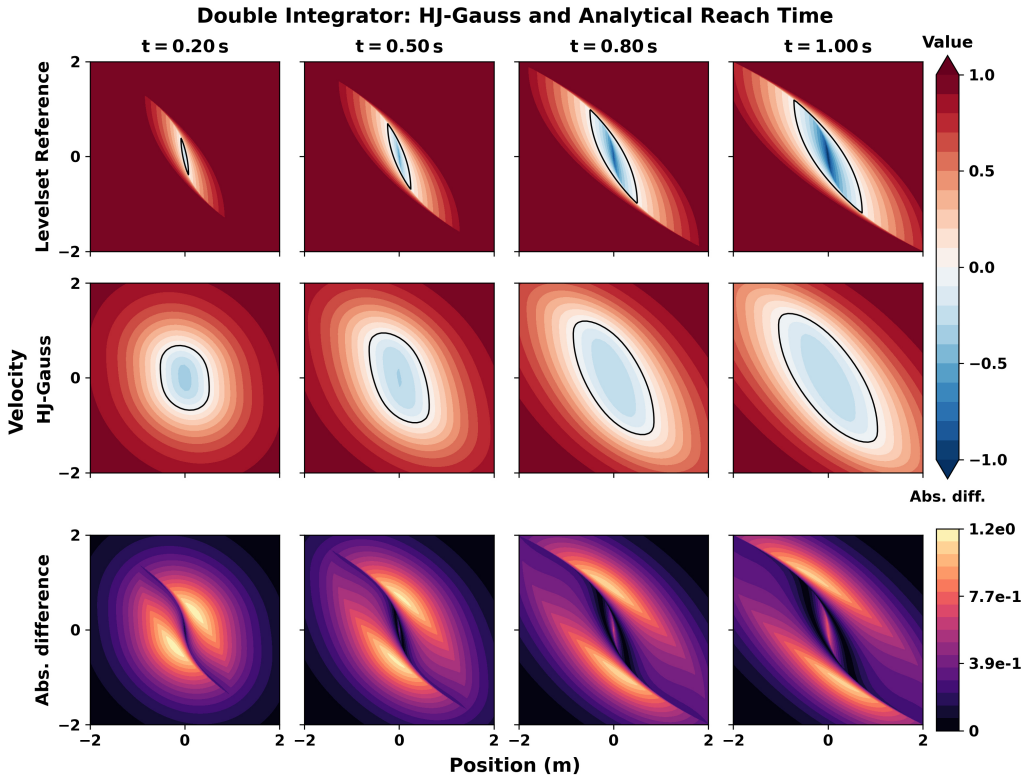


Figure 6: Double-integrator values at four elapsed times with $\delta = 0.08$. Rows show the analytical reference, transformed-solver value, and absolute difference. Black contours mark the zero level sets in the first two rows.

(e) Dubins Vehicles Homicidal Chauffeur Problem

Merz's pursuit–evasion game extends the homicidal chauffeur problem to two vehicles with comparable speeds and turn radii [14, 17]. The experiment computes terminal-horizon viscosity values. Let $\boldsymbol{x} = (x_1, x_2, x_3)$ be the pursuer's position and heading relative to an evader centered at the origin. With constant linear speeds $v_e, v_p > 0$ and steering rates $w_e, w_p \in [-w, w]$, the relative dynamics are [19]

$$\dot{x}_1 = -v_e + v_p \cos x_3 + w_e x_2, \quad \dot{x}_2 = v_p \sin x_3 - w_e x_1, \quad \dot{x}_3 = w_p - w_e. \tag{3.7}$$

Set $v_e = v_p = w = 1$ so that for matched headings, identical steering signals preserve the initial separation. The capture target surface is the cylinder $\mathcal{T} = \{\boldsymbol{x} : x_1^2 + x_2^2 \le r^2\}$, with capture radius $r > 0$. In absolute coordinates, the dynamics is $\dot{x}_1 = \cos\theta$, $\dot{x}_2 = \sin\theta$, and $\dot{\theta} = u$, with $|u| \le 1$ and a pose target that includes heading. Write the Hamiltonian as $\boldsymbol{H} = -p_1 \cos\theta - p_2 \sin\theta + |p_3|$. The physical-time Hamiltonian $\boldsymbol{H}_{\text{phys}} = \max_{w_e} \min_{w_p} p \cdot f$ is

$$\boldsymbol{H}_{\text{phys}}(\boldsymbol{x}, p) = p_1(v_p \cos x_3 - v_e) + p_2(v_p \sin x_3) + w|p_1 x_2 - p_2 x_1 - p_3| - w|p_3|. \tag{3.8}$$

For the normalized bounds $v_e = v_p = w = 1$, the Hamiltonian becomes

$$\boldsymbol{H}_{\text{phys}}(\boldsymbol{x}, p) = p_1(\cos x_3 - 1) + p_2(\sin x_3) + |p_1 x_2 - p_2 x_1 - p_3| - |p_3|. \tag{3.9}$$

For elapsed backward time, the numerical Cauchy problem uses $\boldsymbol{H} = -\boldsymbol{H}_{\text{phys}}$. For unit bounds this is

$$\boldsymbol{H}(\boldsymbol{x}, p) = p_1(1 - \cos x_3) - p_2 \sin x_3 - |p_1 x_2 - p_2 x_1 - p_3| + |p_3|. \tag{3.10}$$

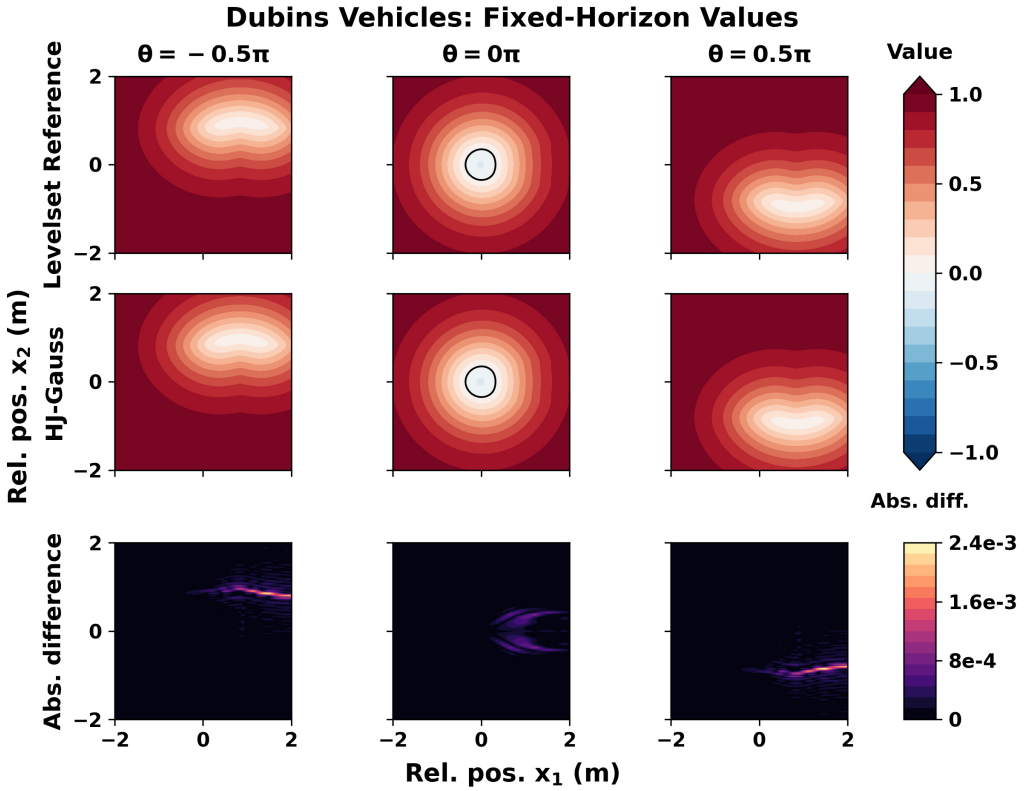


Figure 7: Relative-Dubins terminal-horizon values at $T = 1$ and $\delta = 0.08$. Rows show the `Levelset` Reference [21], `HJ-Gauss`, and their absolute difference. Columns select three headings. Black contours mark the zero level set. The comparison uses the same Hamiltonian, initial condition, and viscosity in both solvers.

The recomputed terminal-horizon fields use this Hamiltonian without a stopping operator.

The levelset comparison [21] with `HJ-Gauss` for the Dubins vehicles in relative coordinates is shown in Figure 7; Table 2 reports its refinement checks.

(f) Numerical setup notes and hyperparameters

Algorithm 1 summarizes the implemented numerical procedure. For the rockets and relative-Dubins cases, the transformed solver uses second-order exponential Runge–Kutta time stepping on a periodic Fourier grid. A fourth-order finite-difference discretization with classical fourth-order Runge–Kutta integration evolves the original viscous equation. The starling-inspired case uses fourth-order centered differences and classical fourth-order Runge–Kutta integration for the transformed equation. Its reference uses sixth-order centered differences and fourth-order Runge–Kutta integration.

All figures assign the `levelset`scales $L_* = 1\,\mathrm{m}$ and $T_* = 1\,\mathrm{s}$, so that positions are displayed in meters and elapsed time in seconds. The Starlings example payoff takes the minimum of seven translated radial payoffs, with target centers on a radius-1.15 ring and disk radius 0.6. The baseline spatial box is $[-6, 6]^2$, with periodic heading $[-\pi, \pi)$ for three-dimensional cases. Completion makes the initial condition equal to 1 near each spatial seam while leaving it unchanged on

Table 1: Maximum sampled changes under time halving, coefficient variation, isolated boundary extension, and combined box sensitivity. Boundary extension fixes the `Levelset` Reference time step, spacing, heading count, Hamiltonian, and initial condition on the common box.

System	Time	Coefficient	Combined box
Rockets	7.19×10^{-6}	1.84×10^{-4}	0.00136
Dubins vehicles	3.97×10^{-5}	2.79×10^{-5}	5.26×10^{-5}
Single Dubins	1.47×10^{-6}	1.02×10^{-5}	8.83×10^{-6}
Double integrator	6.85×10^{-7}	1.73×10^{-6}	4.40×10^{-5}
Bird model	6.23×10^{-7}	7.46×10^{-6}	5.46×10^{-4}

$[-4.5, 4.5]^2$. The observation box is $[-2, 2]^2$ for the single-Dubins, relative-Dubins, and double-integrator cases and $[-3, 3]^2$ for rockets and birds. The prescribed coefficient is

$$\boldsymbol{c}(t, \boldsymbol{x}) = b + 0.04 \cos t \prod_{j=1}^{d} \cos(2\pi \boldsymbol{x}_j / L_j), \qquad b = 0.2, \tag{3.11}$$

where L_j is the corresponding periodic length. The computation includes its time derivative, gradient, and Laplacian. The sensitivity calculation changes b to 0.35 while keeping the Hamiltonian and initial condition fixed.

Successive refinements can change both spatial resolution and time step; fixed-grid time halving is checked separately. Refinement and solver discrepancies are measured on 65×65 observation points and, in three dimensions, 32 equally spaced headings. These are maximum sampled discrepancies of interpolated fields. Equation residuals use native grid states and spectral spatial derivatives and centered differences at one quarter, one half, and three quarters of the elapsed horizon.

The acceptance thresholds were fixed before the refinement cycle: 0.01 for successive field refinements, 0.02 for the levelset reference [21], 0.002 for time halving and coefficient changes, and 0.005 for each domain-extension test. Sampled original-equation residuals must have maximum at most 0.15 and root-mean-square at most 0.02. The box experiment extends the half-width to 7.5 at fixed spatial spacing and heading resolution; it also changes the time step and coefficient wavelength, so it measures their combined sensitivity. A separate boundary-sensitivity test evolves the original equation on the enlarged box while fixing the time step, spatial spacing, heading count, Hamiltonian, and initial condition on the common domain. This separates boundary sensitivity from changes in the transformation coefficient and time step. Diffusion has infinite propagation speed, so neither finite extension supplies a whole-space boundary-error certificate.

Table 1 separates time-step, coefficient, and domain sensitivity. All comparisons meet their thresholds; these sampled checks do not establish the source iteration's invariant-set hypotheses.

(g) Numerical comparison summary

4. Conclusion

The variable-coefficient Cole–Hopf transformation maps the viscous Hamilton–Jacobi equation to a semilinear heat equation with a complete nonlinear source. Retaining the algebraic and coefficient-derivative terms preserves the original regularized equation under positive logarithmic inversion. Deterministic solvers integrate the transformed field by Fourier collocation or centered finite differences. The nonlinear Monte Carlo Picard solver instead evaluates individual states through nested Gaussian value-and-gradient updates.

The state-dimension sweep supports the narrow storage advantage of this point evaluator. At fixed sampling and depth, dimensions 8–2048 require no tensor grid and have a counted

Table 2: Maximum sampled discrepancies for the `HJ-Gauss` computations. Refinement and `Levelset` Reference columns use the common observation queries; the residual column gives the largest sampled residual over native states. The grid column lists the fine spatial and heading counts.

System	Grid	Refinement	Levelset Reference	Residual
Rockets	$160^2 \times 80$	0.007481	0.003223	0.1433
Dubins vehicles	$224^2 \times 112$	0.005717	0.002511	0.1379
Single Dubins	$128^2 \times 64$	0.009297	0.001415	0.02992
Double integrator	128^2	0.006741	5.60×10^{-4}	0.01187
Bird model	$192^2 \times 96$	0.008817	6.44×10^{-4}	0.1295

coordinate payload linear in dimension. The sample sweep gives a different runtime result: increasing samples per update from 4 to 48 raises median runtime from 0.0052 to 0.9355 seconds. The nested implementation therefore does not establish $O(Nn)$ end-to-end runtime or favorable scaling with Picard depth.

Analytic solutions, refinement studies, coefficient variation, domain extension, and same-viscosity reference fields test the implementation. The rockets, relative-Dubins, single-Dubins, double-integrator, and starling-inspired cases yield maximum sampled reference discrepancies below 3.3×10^{-3} on the reported observation queries. These comparisons validate the stated discretizations on finite periodic boxes. They do not establish whole-space error bounds, dimension-independent complexity, or continuous-state safety certificates.

The 100,000-state experiment evaluates one precomputed three-dimensional relative-state field at independently sampled four-component queries. It demonstrates repeated interpolation of a reduced value field; it does not solve a coupled flock game. The associated component, hole, and Euler counts describe thresholded numerical slices. They provide useful geometric diagnostics without implying biological transition laws or invariant barriers.

The source code records solver settings, fresh-process scaling replicates, operation counts, refinement checks, residual samples, and archive provenance. The Monte Carlo implementation rejects nonpositive transformed estimates and reports finite Picard levels rather than asserting fixed-point convergence. A continuum reachability result would still require controlled field reconstruction, accumulated sampling-error bounds, verified admissibility, and a stopping rule tied to the contraction hypotheses.

References

1 M. Althoff, G. Frehse, and A. Girard. Set Propagation Techniques for Reachability Analysis. *Annual Review of Control, Robotics, and Autonomous Systems*, 4:369–395, 2021. ISSN 25735144.

2 J.-P. Aubin, A. M. Bayen, and P. Saint-Pierre. *Viability Theory*. Number August 2015. 2011. ISBN 9783642166846.

3 S. P. Bhat and D. S. Bernstein. Continuous finite-time stabilization of the translational and rotational double integrators. *IEEE Transactions on automatic control*, 43(5):678–682, 1998.

4 M. G. Crandall and P.-L. Lions. Viscosity solutions of hamilton-jacobi equations. *Transactions of the American mathematical society*, 277(1):1–42, 1983.

5 M. G. Crandall and P.-L. Lions. Two Approximations of Solutions of Hamilton-Jacobi Equations. *Mathematics of Computation*, 43(167):1–19, 1984.

6 M. G. Crandall and A. Majda. Monotone Difference Approximations For Scalar Conservation Laws. *Mathematics of Computation*, 34(149):1–21, 1980.

7 M. G. Crandall, L. C. Evans, and P. L. Lions. Some Properties of Viscosity Solutions of Hamilton-Jacobi Equations. *Transactions of the American Mathematical Society*, 282(2):487, 1984. ISSN 00029947.

8 Defense Acquisition University. Verification, 2023. URL https://www.dau.edu/tools/se-brainbook/Pages/Technical%20Processes/verification.aspx. Accessed April 5, 2023.

9 S. E. Dreyfus. Control Problems With Linear Dynamics, Quadratic Criterion, and Linear Terminal Constraints. Technical report, Rand Corp, Santa Monica Calif, 1966.

10 L. Evans and P. E. Souganidis. Differential Games And Representation Formulas For Solutions Of Hamilton-Jacobi-Isaacs Equations. *Indiana Univ. Math. J*, 33(5):773–797, 1984. ISSN 0022-2518.

11 K. L. Hobbs, M. L. Mote, M. C. Abate, S. D. Coogan, and E. M. Feron. Runtime assurance for safety-critical systems. *IEEE Control Systems Magazine*, 43:28–65, 2023.

12 M. Hutzenthaler and T. Kruse. Multilevel Picard Approximations of High-Dimensional Semilinear Parabolic Differential Equations with Gradient-Dependent Nonlinearities. *SIAM Journal on Numerical Analysis*, 58(2):929–961, 2020. doi: 10.1137/17M1157015.

13 R. Isaacs. Differential Games: A Mathematical Theory with Applications to Warfare and Pursuit, Control and Optimization., 1966.

14 R. Isaacs. *Differential Games: A Mathematical Theory with Applications to Warfare and Pursuit, Control and Optimization.* Kreiger, Huntigton, NY, 1999.

15 C. Livadas and N. A. Lynch. Formal verification of safety-critical hybrid systems. In *International Workshop on Hybrid Systems: Computation and Control*, pages 253–272. Springer, 1998.

16 J. Lygeros. On reachability and minimum cost optimal control. *Automatica*, 40(6):917–927, 2004.

17 A. Merz. The game of two identical cars. *Journal of Optimization Theory and Applications*, 9(5): 324–343, 1972.

18 I. Mitchell. A Robust Controlled Backward Reach Tube with (Almost) Analytic Solution for Two Dubins Cars. *EPiC Series in Computing*, 74:242–258, 2020.

19 I. M. Mitchell, A. M. Bayen, and C. J. Tomlin. A Time-Dependent Hamilton-Jacobi Formulation of Reachable Sets for Continuous Dynamic Games. *IEEE Transactions on Automatic Control*, 50 (7):947–957, 2005. ISSN 00189286.

20 L. Molu. The Python LevelSet Toolbox (LevelSetPy). In *2024 IEEE 63rd Conference on Decision and Control (CDC)*, pages 8938–8945, 2024. doi: 10.1109/CDC56724.2024.10886640.

21 L. Molu. LevelSetPy: A GPU-Accelerated Package for Hyperbolic Hamilton-Jacobi Partial Differential Equations' Solubility. *The ACM Transactions on Mathematical Software*, 2025.

22 S. Osher and R. Fedkiw. Level Set Methods and Dynamic Implicit Surfaces. *Applied Mechanics Reviews*, 57(3):B15–B15, 2004. ISSN 0003-6900.

23 S. Osher and J. A. Sethian. Fronts Propagating with Curvature-Dependent Speed: Algorithms based on Hamilton-Jacobi Formulations. *Journal of Computational Physics*, 79(1):12–49, 1988. ISSN 10902716.

24 S. Osher and C.-W. Shu. High-Order Essentially Nonoscillatory Schemes for Hamilton-Jacobi Equations. *SIAM Journal of Numerical Analysis*, 28(4):907–922, 1991. ISSN 15333868.

25 C. J. Tomlin, J. Lygeros, and S. S. Sastry. A game theoretic approach to controller design for hybrid systems. *Proceedings of the IEEE*, 88(7):949–970, 2000.

26 W. M. Wonham. Linear multivariable control: A geometric approach. *Applications of Mathematics*, 10, 1985.

Appendices

A. Heat-kernel and potential representations

The transformed equation separates diffusion from a nonlinear source. For a prescribed bounded source f, the heat semigroup gives

$$z(t,\boldsymbol{x}) = \mathbb{E}[z_0(\boldsymbol{x} + \sqrt{\delta t}\, Z)] + \int_0^t \mathbb{E}[f(s, \boldsymbol{x} + \sqrt{\delta(t-s)}\, Z)]\, ds, \qquad Z \sim \mathcal{N}(0, I_n). \tag{A.1}$$

This representation explains the diffusion operator used by the exponential integrator. The reported numerical solvers evaluate the field on deterministic grids; they do not estimate (A.1) by random sampling.

For $t > 0$, differentiation of the heat kernel yields

$$Dz(t,\boldsymbol{x}) = \frac{1}{\sqrt{\delta t}}\mathbb{E}[Z z_0(\boldsymbol{x}+\sqrt{\delta t}\,Z)] + \int_0^t \frac{\mathbb{E}[Z f(s,\boldsymbol{x}+\sqrt{\delta(t-s)}\,Z)]}{\sqrt{\delta(t-s)}}\,ds. \tag{A.2}$$

The singularity at $s = t$ is integrable in the deterministic semigroup bound used in Appendix B. Spectral or finite-difference differentiation supplies the gradients in the experiments.

A prescribed bounded continuous potential R gives a related Feynman–Kac representation. With $X_r = \boldsymbol{x} + \sqrt{\delta}W_r$ for Brownian motion W, the solution of $z_t = \delta\Delta z/2 + Rz$ satisfies

$$z(t,\boldsymbol{x}) = \mathbb{E}_{\boldsymbol{x}}\left[z_0(X_t)\exp\left\{\int_0^t R(t-r,X_r)\,dr\right\}\right]. \tag{A.3}$$

When R depends on the unknown value and gradient, this formula remains implicit. The deterministic solvers therefore integrate the complete additive source in (2.10).

B. Contraction, residual bounds, and robustness

This appendix gives conditional convergence and a residual certificate for the full transformed equation.

(a) Convergence of source iteration

Gradient dependence requires a norm that controls both values and first derivatives. Let $\mathrm{BUC}^1(\mathbb{R}^n)$ contain continuously differentiable functions with bounded, uniformly continuous values and gradients. On $X_\tau = C([0,\tau];\mathrm{BUC}^1)$, use

$$\|z\|_{X_\tau} = \sup_{s\le\tau}\|z(s)\|_\infty + \sup_{s\le\tau}\|Dz(s)\|_\infty. \tag{B.1}$$

For $\boldsymbol{g} \in \mathrm{BUC}^1$ and bounded coefficient derivatives, the transformed initial condition belongs to the same space. Define the mild update by

$$(\mathcal{M}z)(t) = S_t\boldsymbol{\omega}_0 + \int_0^t S_{t-s}F(s,\cdot,z(s),Dz(s))\,ds. \tag{B.2}$$

Theorem B.1 (Conditional finite-time convergence)**.** *Let $\mathcal{A} \subset X_\tau$ be nonempty, closed, and invariant under $\mathcal{M}$. Suppose all its members satisfy $a \le z \le b$ and $|Dz| \le Q$, with $a > 0$. Assume that the source is bounded and uniformly continuous on the relevant coefficient and state ranges, and satisfies*

$$|F(t,\boldsymbol{x},z,q) - F(t,\boldsymbol{x},\widetilde{z},\widetilde{q})| \le L\big(|z-\widetilde{z}| + |q-\widetilde{q}|\big). \tag{B.3}$$

If $\theta = L(\tau + 2\sqrt{n\tau/\delta}) < 1$, then $\mathcal{M}$ has a unique fixed point in $\mathcal{A}$. Its iterates satisfy

$$\|z^{(K)} - \boldsymbol{\omega}\|_{X_\tau} \le \theta^K\|z^{(0)} - \boldsymbol{\omega}\|_{X_\tau}, \qquad \|z^{(K)} - \boldsymbol{\omega}\|_{X_\tau} \le \frac{\theta}{1-\theta}\|z^{(K)} - z^{(K-1)}\|_{X_\tau} \tag{B.4}$$

for $K \ge 1$. Here finite-time convergence means convergence as $K \to \infty$ on the finite elapsed-time interval $[0,\tau]$; it does not assert termination after finitely many updates. If the fixed point is classical, its inverse solves the original viscous equation.

Proof of Theorems 2.1 and B.1. The space $\mathrm{BUC}^1(\mathbb{R}^n)$ is Banach under the sum of its uniform value and gradient norms. Hence X_τ is Banach, and its closed nonempty subset $\mathcal{A}$ is complete. Invariance makes $\mathcal{M}$ a self-map on this set.

For $z, \widetilde{z} \in \mathcal{A}$, write

$$d(s,\boldsymbol{x}) = F(s,\boldsymbol{x},z(s,\boldsymbol{x}),Dz(s,\boldsymbol{x})) - F(s,\boldsymbol{x},\widetilde{z}(s,\boldsymbol{x}),D\widetilde{z}(s,\boldsymbol{x})).$$

The source Lipschitz bound gives $\|d(s)\|_\infty \le L\|z-\widetilde{z}\|_{X_\tau}$ for every $s \in [0,\tau]$. The common initial-condition terms cancel in $\mathcal{M}z - \mathcal{M}\widetilde{z}$. Since the heat semigroup contracts the uniform norm,

$$\|\mathcal{M}z(t) - \mathcal{M}\widetilde{z}(t)\|_\infty \le \int_0^t \|d(s)\|_\infty \, ds \le Lt\|z-\widetilde{z}\|_{X_\tau}. \tag{B.5}$$

Gaussian differentiation, followed by $\mathbb{E}|Z| \le \sqrt{n}$ for $Z \sim \mathcal{N}(0, I_n)$, yields

$$\|DS_r h\|_\infty \le \frac{\mathbb{E}|Z|}{\sqrt{\delta r}}\|h\|_\infty \le \sqrt{\frac{n}{\delta r}}\|h\|_\infty, \qquad r > 0. \tag{B.6}$$

The singularity at $r = 0$ is integrable. Differentiating the source integral therefore gives

$$\|D\mathcal{M}z(t) - D\mathcal{M}\widetilde{z}(t)\|_\infty \le L\int_0^t \sqrt{\frac{n}{\delta(t-s)}}\, ds\, \|z-\widetilde{z}\|_{X_\tau} = 2L\sqrt{\frac{nt}{\delta}}\|z-\widetilde{z}\|_{X_\tau}. \tag{B.7}$$

Taking the two time suprema and adding proves

$$\|\mathcal{M}z - \mathcal{M}\widetilde{z}\|_{X_\tau} \le L\left(\tau + 2\sqrt{n\tau/\delta}\right)\|z-\widetilde{z}\|_{X_\tau} = \theta\|z-\widetilde{z}\|_{X_\tau}.$$

The assumption $\theta < 1$ and Banach's fixed-point theorem give a unique fixed point $\boldsymbol{\omega} \in \mathcal{A}$. For each starting iterate $z^{(0)} \in \mathcal{A}$, invariance keeps every subsequent iterate in $\mathcal{A}$. Repeated contraction against the fixed point gives the first estimate in (B.4).

For the posterior estimate, contraction between successive iterates gives

$$\|z^{(K+j+1)} - z^{(K+j)}\|_{X_\tau} \le \theta^{j+1}\|z^{(K)} - z^{(K-1)}\|_{X_\tau}, \qquad j \ge 0.$$

Sum these differences from $j = 0$ to infinity and use convergence to the fixed point. The geometric series has sum $\theta/(1-\theta)$, proving the second estimate. When $\theta = 0$, the map is constant on $\mathcal{A}$ and one update reaches its fixed point.

Finally, $\boldsymbol{\omega} \ge a > 0$ permits logarithmic inversion. If the fixed point is classical, its mild equation implies the full transformed differential equation. Lemma 2.1 then shows that its inverse solves the original viscous equation. □

For bounded coefficient derivatives, $\boldsymbol{c} \ge \boldsymbol{c}_- > 0$, and $z \ge a > 0$, the inverse maps have bounded derivatives on bounded (z,q) ranges. A Hamiltonian uniformly bounded and locally Lipschitz in p on those ranges, uniformly over $(t,\boldsymbol{x})$, therefore supplies a finite source constant L. No convexity assumption is needed for this contraction argument.

The global complexity theorem in [12, Corollary 4.8] does not follow from this local argument. Their setting imposes a globally Lipschitz source in the value and gradient variables, together with strong smoothness and derivative-growth bounds for the complexity result. The transformed source need not meet those assumptions. Even with constant $\boldsymbol{c}$ and the linear Hamiltonian $\boldsymbol{H}(p) = ap$ in one dimension, it becomes

$$F(z,q) = -aq - \frac{\delta}{2z}q^2. \tag{B.8}$$

This source is singular as z approaches zero and is quadratic in q. Using a globally Lipschitz extension to analyze the original equation would require a proof that the solution remains inside the bounded positive region on which the sources agree. Transferring their complexity result would also require verifying their dimension-dependent constants and quadrature regularity. The present contraction theorem establishes neither requirement.

Invariance is a substantive hypothesis. One sufficient construction starts with $b_0(t) = S_t\boldsymbol{\omega}_0$ and an X_τ ball of radius r about b_0. Choose $r < \inf \boldsymbol{\omega}_0$ to preserve positivity. If $|F| \le B_F$ throughout

that ball and

$$B_F\left(\tau + 2\sqrt{n\tau/\delta}\right) \le r, \tag{B.9}$$

then the mild map preserves the ball. Under the preceding uniform-in-state Hamiltonian and coefficient assumptions, bounded initial values and gradients give finite source ranges for sufficiently small τ. The case of a constant $\boldsymbol{c} = 1/\delta$ and $\boldsymbol{H} = |p|^2/2$ has $F = 0$ and $\theta = 0$, so the hypotheses include a nonempty solvable case. Extension to a longer horizon requires renewed bounds and invariance on each interval; a global contraction is not asserted.

If $0 < \theta < 1$ and the initial error is bounded by $E_0 > 0$, a sufficient iteration count is

$$K \ge \max\left\{0, \left\lceil \frac{\log(E_0/\varepsilon)}{\log(1/\theta)} \right\rceil \right\}. \tag{B.10}$$

When $\theta = 0$, one update reaches the fixed point. A small change between noisy iterates alone does not certify convergence.

(b) Scope of the deterministic implementation

The contraction theorem concerns the whole-interval mild map. The reported solvers instead apply exponential or classical Runge–Kutta time integration after spatial discretization. Their refinement studies measure discretization consistency on the stated periodic boxes. They do not verify the invariant-set hypothesis of Theorem B.1.

The implementation checks positivity at every time step and evaluates equation residuals at selected native-grid states and times. These checks detect numerical failures and support comparisons across refinements. They do not bound the continuum residual between grid states or between sampled times.

(c) A posteriori certificate for the original equation

A residual of the full transformed equation measures the residual of the original equation directly. Assume comparison for the original parabolic equation on the whole space. Let U be a bounded classical approximation with bounded residual, and define

$$w = e^{-\boldsymbol{c}U}, \qquad d = w_t - \frac{\delta}{2}\Delta w - F(t, \boldsymbol{x}, w, Dw). \tag{B.11}$$

Theorem B.2 (Residual certificate)**.** *Under these hypotheses, the viscous solution with initial condition* $\boldsymbol{g}$ *obeys*

$$\|U(t) - \boldsymbol{v}^\delta(t)\|_\infty \le \|U(0) - \boldsymbol{g}\|_\infty + \int_0^t \left\| \frac{d(s)}{\boldsymbol{c}(s)w(s)} \right\|_\infty ds. \tag{B.12}$$

Proof. Lemma 2.1 gives $d = -\boldsymbol{c}wr_U$, with $r_U = U_t + \boldsymbol{H}(t, \boldsymbol{x}, DU) - \delta\Delta U/2$. Let $e_0 = \|U(0) - \boldsymbol{g}\|_\infty$ and $E(t) = e_0 + \int_0^t \|r_U(s)\|_\infty ds$. Since the Hamiltonian does not depend on the value, $U - E$ is a subsolution and $U + E$ is a supersolution. Their initial conditions bracket $\boldsymbol{g}$. Comparison gives $U - E \le \boldsymbol{v}^\delta \le U + E$, proving the bound. □

The source must include both residual components when computing d. If a solver omits $\boldsymbol{R}_{\mathrm{der}}$, then its transformed residual contains the missing term $-w\boldsymbol{R}_{\mathrm{der}}$. The certificate charges that omission. Residuals evaluated only at sampled states are diagnostic; rigorous use requires a bound between those states and through time, including derivative approximation error.

(d) Viscosity, model perturbations, and level sets

Vanishing viscosity remains a separate approximation. Suppose the Hamiltonian and initial condition satisfy a comparison and approximation theorem giving $\|\boldsymbol{v}^\delta - \boldsymbol{v}^0\|_\infty \le C_{\mathrm{vis}}\sqrt{\delta}$ on

the chosen horizon. This rate is an additional hypothesis, rather than a consequence of the transformation. Under the same uniform norm and domain, Theorem B.2 then gives

$$\|U(t) - \boldsymbol{v}^0(t)\|_\infty \le \|U(0) - \boldsymbol{g}\|_\infty + \int_0^t \left\| \frac{d(s)}{\boldsymbol{c}(s)w(s)} \right\|_\infty ds + C_{\mathrm{vis}}\sqrt{\delta}. \tag{B.13}$$

Proposition B.1 (Hamiltonian and initial condition perturbations)**.** *Suppose two equations with the same diffusion admit comparison, their initial conditions differ by at most ϵ_g, and $|\boldsymbol{H}_1(t,\boldsymbol{x},p) - \boldsymbol{H}_2(t,\boldsymbol{x},p)| \le \epsilon_H$ throughout the required co-state range. For classical solutions, it suffices that this range contains $D\boldsymbol{v}_2$. Then*

$$\|\boldsymbol{v}_1(t) - \boldsymbol{v}_2(t)\|_\infty \le \epsilon_g + t\epsilon_H. \tag{B.14}$$

Proof. The functions $\boldsymbol{v}_2 - \epsilon_g - t\epsilon_H$ and $\boldsymbol{v}_2 + \epsilon_g + t\epsilon_H$ are a subsolution and a supersolution for the first equation. Comparison proves the claim. For viscosity solutions, impose the Hamiltonian bound on the test-gradient range required by comparison, or uniformly over all p. □

Changing a prescribed positive transformation coefficient does not perturb the Hamiltonian when both source components are retained. Thus a smooth positive regularization that preserves the stated differentiability changes coordinates; it creates no Hamiltonian modeling error by itself. Hard clipping can destroy that differentiability and falls outside these classical results. If the Hamiltonian is deliberately replaced, the perturbation bound applies to that separate change. The matching choice $\boldsymbol{c} = 2\boldsymbol{H}(t,\boldsymbol{x},D\boldsymbol{v})/(\delta|D\boldsymbol{v}|^2)$ can vanish, change sign, or become singular. At $\boldsymbol{c} = 0$, the exponential transform loses the value information. Recomputing this ratio also requires regularity beyond the prescribed-coefficient hypotheses. `HJ-Gauss` permits a positive constant coefficient even in these cases.

If a uniform bound $|U - \boldsymbol{v}^0| \le E$ has been established on a specified domain, then

$$\{U < -E\} \subseteq \{\boldsymbol{v}^0 < 0\}, \qquad \{U > E\} \subseteq \{\boldsymbol{v}^0 > 0\}. \tag{B.15}$$

Proof of Corollary 2.3. Fix a point in the specified set and suppress its arguments. The assumed error bound is equivalent to

$$U - E \le \boldsymbol{v}^0 \le U + E. \tag{B.16}$$

If $U > E$, then the lower bound gives $\boldsymbol{v}^0 \ge U - E > 0$. Thus the point lies outside the target-reaching class $\{\boldsymbol{v}^0 \le 0\}$. If $U < -E$, then the upper bound gives $\boldsymbol{v}^0 \le U + E < 0$, placing the point in the strict negative sublevel set. Since the argument applies at every point covered by the error bound, it proves both inclusions in (B.15).

The states assigned neither strict sign by this rule satisfy $-E \le U \le E$, which is the abstention band $|U| \le E$. At its boundaries, the corresponding estimate in (B.16) is nonstrict, so the corollary makes no strict sign assertion there.

If the assumed error bound holds simultaneously on an event of probability at least $1 - \eta$, both sign implications hold simultaneously on that same event. No additional union bound is needed. The domain of the sign certificate is the domain on which that error bound has been established; a bound at finitely many queries supplies no assertion between those queries. □

Interpreting either sign as safety also requires the corresponding game's value convention and, for reach–avoid problems, a correctly solved obstacle problem.